

Hierarchical Inheritance and Method Overloading Program

Program Code:

```
class A {
    void smtg() {
        System.out.println("This is Class A");
    }

    void add() {
        int a = 10, b = 20;
        System.out.println("Addition is: " + (a + b));
    }
}

class B extends A {
    void smtg2() {
        System.out.println("This is Class B");
    }

    void add(int c) {
        int a = 10, b = 20;
        System.out.println("Addition is: " + (a + b + c));
    }
}

class C extends A {
    void smtg3() {
        System.out.println("This is Class C");
    }

    void add(int c, int d) {
        int a = 10, b = 20;
        System.out.println("Addition is: " + (a + b + c + d));
    }
}

public class Main {
    public static void main(String[] args) {

        B obj1 = new B();
        C obj2 = new C();

        obj1.smtg();
        obj1.smtg2();
        obj1.add(30);

        obj2.smtg();
        obj2.smtg3();
        obj2.add(30, 40);
    }
}
```

Explanation:

This program demonstrates hierarchical inheritance and method overloading. Classes B and C both extend class A. Object obj1 of class B accesses the inherited method from class A and the method of class B. Object obj2 of class C accesses the inherited method from class A and the method of class C. The add() method is overloaded with different parameters to perform addition.

Compilation and Execution Commands:

Save the file as: Main.java

Compile:
javac Main.java

Run:
java Main

Output Screenshot:

```
This is Class A  
This is Class B  
Addition is: 60  
This is Class A  
This is Class C  
Addition is: 100
```

Verification:

The output confirms that both class B and class C inherit the smtg() method from class A. Class B displays its own method and calculates 60 using add(30). Class C displays its own method and calculates 100 using add(30, 40).

Result:

Thus, the Java program was successfully executed and hierarchical inheritance with method overloading was verified successfully.